二维数组练习2

人员

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上周作业检查

上周作业链接: https://cppoj.kids123code.com/contest/3840

比赛概况 题目列表 选择题列表 提交记录 实时榜单 选择题排行榜 编辑比赛

王向东老师周六三点半C++二维数组练习1

刷新

#	用户名	姓名	编程分	时间	A	B	C	D	E	F	G	H
1	zhaotingshu	赵庭澍	800	20	100	100	100	100	100	100	100	100
2	wangchunshuo	王纯硕	800	22	100	100	100	100	100	100	100	100
3	lihonghao	李弘浩	800	24	100	100	100	100	100	100	100	100
4	chenyiqi	陈奕圻	800	26	100	100	100	100	100	100	100	100
5	cuixiaoze	崔孝泽	800	26	100	100	100	100	100	100	100	100
6	quyuxuan	屈钰轩	800	28	100	100	100	100	100	100	100	100
7	pengjunwen	彭俊文	700	23	100	100	100	100	100	100	100	
8	xiejuntian	解钧添	700	26	100	100	100	100	100	100	100	
9	zhangjiarui	张嘉芮	700	28	100	100	100	100	100	100	0	100
10	maxinyi	马欣怡	618	18	100	100	100	100	100	100	18	
11	liuchenxi123	刘晨锡	500	11	100	100	100	100	100			

作业

https://cppoj.kids123code.com/contest/4030 (课上讲了 A ~ E 题, 作业是 A ~ E 题必做)

课堂表现

今天的第 5 题比较复杂一些, 同学们课上基本都没有做完这道题, 课下要再好好复盘一下这道题目。

课堂内容

两个数相邻吗?

```
#include<bits/stdc++.h>
using namespace std;
int n,m;
int a[105][105];
int c,b;
int bx,by,cx,cy;
int main()
{
    cin>>n>>m;
    for(int i=1;i<=n;i++)
```

```
    {
        for(int j=1;j<=m;j++)
        {
            cin>>a[i][j];
        }
    }
    cin>>b>>c;
    for(int i=1;i<=n;i++)
    {
        for(int j=1;j<=m;j++)
        {
            if(a[i][j]==b)
            {
                bx=i;
                by=j;
            }
            if(a[i][j]==c)
            {
                cx=i;
                cy=j;
            }

        }
    }
    if((bx==cx&&abs(by-cy)==1)||(by==cy&&abs(cx-bx)==1))
    {
        cout<<"Y";
    }
    else cout<<"N";
}
```

地雷数量求解

```
#include <bits/stdc++.h>
using namespace std;
int a[205][205];
int cnt;
int main()
{
    int n,m;
    cin>>n>>m;
    for(int i=1;i<=n;i++)
    {
        for(int j=1;j<=m;j++)
        {
            cin>>a[i][j];
            int x=0;
            while(a[i][j]>0)
            {
                x+=a[i][j]%10;
                a[i][j]/=10;
```

```
            }
            if(x%2!=0)
            {
                cnt++;
            }
        }
    }
    cout<<cnt;
}
```

计算鞍点

```
#include <bits/stdc++.h>

using namespace std;

int a[15][15];

int main()
{
    for (int i = 1; i <= 5; i++) {
        for (int j = 1; j <= 5; j++) {
            cin >> a[i][j];
        }
    }
    int cnt = 0, maxx, minn;
    for (int i = 1; i <= 5; i++) {
        for (int j = 1; j <= 5; j++) {
            maxx = 0;
            for (int k = 1; k <= 5; k++) {
                if (a[i][k] > maxx) {
                    maxx = a[i][k];
                }
            }
            minn = 9999999;
            for (int k = 1; k <= 5; k++) {
                if (a[k][j] < minn) {
                    minn = a[k][j];
                }
            }
            if (a[i][j]==maxx && a[i][j]==minn) {
                cout << i << " " << j << " " << a[i][j] << endl;
                cnt++;
            }
        }
    }
    if (cnt == 0) {
        cout << "not found" << endl;
    }
```

```
    return 0;
}
```

图像相似度

```
#include <bits/stdc++.h>
using namespace std;
int a[205][205],b[205][205];
double x;
int main()
{
    int n,m;
    cin>>n>>m;
    for(int i=1;i<=n;i++)
    {
        for(int j=1;j<=m;j++)
        {
            cin>>a[i][j];
        }
    }
    for(int i=1;i<=n;i++)
    {
        for(int j=1;j<=m;j++)
        {
            cin>>b[i][j];
        }
    }
    for(int i=1;i<=n;i++)
    {
        for(int j=1;j<=m;j++)
        {
            if(a[i][j]==b[i][j])
            {
                x++;
            }
        }
    }
    cout<<fixed<<setprecision(2)<<1.0*x/(n*m)*100;
}
```

孤独的素数

```
#include <bits/stdc++.h>

using namespace std;

int a[55][55], b[55][55];
```

```
int main()
{
    int n, m;
    cin >> n >> m;
    for (int i = 1; i <= n; i++) {
        for (int j = 1; j <= m; j++) {
            cin >> a[i][j];
        }
    }

    for (int i = 1; i <= n; i++) {
        for (int j = 1; j <= m; j++) {
            int cnt = 0;
            for (int k = 1; k <= a[i][j]; k++) {
                if (a[i][j]%k == 0) {
                    cnt++;
                }
            }
            if (cnt == 2) {
                b[i][j] = 1;
            }
            else {
                b[i][j] = 0;
            }
        }
    }

    int cntt = 0;
    for (int i = 1; i <= n; i++) {
        for (int j = 1; j <= m; j++) {
            if (b[i][j] == 1) {
                if (b[i-1][j-1]==0 && b[i-1][j]==0 && b[i-1][j+1]==0 && b[i][j-
1]==0 && b[i][j+1]==0 && b[i+1][j-1]==0 && b[i+1][j]==0 && b[i+1][j+1]==0) {
                    cntt++;
                }
            }
        }
    }
    cout << cntt << endl;
    return 0;
}
```

```
#include <bits/stdc++.h>

using namespace std;

int a[55][55], b[55][55];

int main()
{
    int n, m;
```

```
    cin >> n >> m;
    for (int i = 1; i <= n; i++) {
        for (int j = 1; j <= m; j++) {
            cin >> a[i][j];
        }
    }

    for (int i = 1; i <= n; i++) {
        for (int j = 1; j <= m; j++) {
            int cnt = 0;
            for (int k = 1; k <= a[i][j]; k++) {
                if (a[i][j]%k == 0) {
                    cnt++;
                }
            }
            if (cnt == 2) {
                b[i][j] = 1;
            }
            else {
                b[i][j] = 0;
            }
        }
    }

    int cntt = 0;
    for (int i = 1; i <= n; i++) {
        for (int j = 1; j <= m; j++) {
            if (b[i][j] == 1) {
                int cnt = 0;
                for (int x = i-1; x <= i+1; x++) {
                    for (int y = j-1; y <= j+1; y++) {
                        if (b[x][y] == 1) {
                            cnt++;
                        }
                    }
                }
                if (cnt == 1) {
                    cntt++;
                }
            }
        }
    }
    cout << cntt << endl;
    return 0;
}
```